

optimization-problem

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1 Introduction

Full term list for the DVR optimization problem

v : the verifier assigned to check the draft model's output. Takes one of four values: none, small, medium, large.

k : the draft length, the number of tokens the draft model proposes before they get sent for checking. Takes one of three values: 3, 5, or 7. Only applies when v is not none.

s : the task being routed for. Takes one of three values: Exploration, Reasoning, Conclusion.

(v, k) : a candidate, one specific combination of verifier and draft length, e.g. medium verifier with draft length 5. There are 10 candidates total: none (1), plus small, medium, large each at $k \in \{3, 5, 7\}$ (9).

query: the text of the user's question or prompt that came in.

f_{enc} : the pretrained text encoder. A fixed, pre-trained tool that converts text into a list of numbers (a vector).

$f_{\text{enc}}(\text{query})$: the query, after being converted into a vector by the encoder.

$g(v, k, s)$: the candidate's vector. Built from a short, written description of verifier v , converted into numbers by the same encoder, with k attached as a plain number alongside it.

$\oplus$: the symbol for concatenation. Two lists of numbers joined end to end into one longer list. Not addition.

$z(v, k, s)$: the combined vector, $z(v, k, s) = f_{\text{enc}}(\text{query}) \oplus g(v, k, s)$. The single input fed into the predictor.

MLP: Multi-Layer Perceptron, the trained predictor itself, a small neural network. Takes $z(v, k, s)$ in, produces one raw number out. Its internal weights are learned through training beforehand, not written by hand.

σ (**sigma**): the sigmoid function, $\sigma(x) = 1/(1 + e^{-x})$. Takes any number and squashes it into the range between 0 and 1.

$A(v, k, s)$: the predicted accuracy of candidate (v, k) for task s . $A(v, k, s) = \sigma(\text{MLP}(z(v, k, s)))$. A number between 0 and 1. Higher is better. The only term computed live, per query.

BERTScore: a method for scoring similarity between a model’s output and the correct answer, used to generate training labels for the MLP. Used only during training, not at decision time.

$E(v, k)$: the raw measured energy cost (grams of CO₂) of running verifier v on a draft length of k tokens. Measured once, offline, using CodeCarbon, before any real query is processed.

EnergyNorm (v, k) : $E(v, k)$ rescaled onto a 0 to 1 scale, comparable across all 10 candidates. Higher value means more carbon costly, worse.

$T_{\text{raw}}(v, k)$: the raw measured time (verification time plus transmission time between the A20 and A40 GPUs) for candidate (v, k) . Measured once, offline.

TimeNorm (v, k) : $T_{\text{raw}}(v, k)$ rescaled onto a 0 to 1 scale, comparable across all 10 candidates. Higher value means slower, worse.

w_A, w_E, w_T : three weights, numbers between 0 and 1, summing to 1 together. They control how much each term (Accuracy, Energy, Time) counts toward the final decision.

CombinedScore (v, k, s) : the final number computed for each candidate: $\text{CombinedScore}(v, k, s) = w_A \cdot A(v, k, s) - w_E \cdot \text{EnergyNorm}(v, k) - w_T \cdot \text{TimeNorm}(v, k)$. Accuracy is added; Energy and Time are subtracted.

argmax: means find and return whichever candidate scores the highest. $\arg \max_{(v, k)} \text{CombinedScore}(v, k, s)$ means look at all 10 candidates for task s and pick the one with the biggest number.

$(v^*(s), k^*(s))$: the final answer, the verifier and draft length chosen for task s , the winner of the argmax.

2 Exploration

Part A: Calculating the accuracy term

For each of the 10 candidates (v, k) , the predicted accuracy $A(v, k, \text{Exploration})$ is produced by a four-step calculation:

1. The query is converted into numbers by an encoder. Separately, a written description of verifier v is converted into numbers by the same encoder, and the numeric value of k is attached alongside it. These are joined together into one vector, $z(v, k, \text{Exploration})$.

2. $z(v, k, \text{Exploration})$ is passed through the MLP, a trained small network, producing one raw number.
3. That raw number is passed through σ , squashing it into the range 0 to 1.
4. The result is $A(v, k, \text{Exploration})$, the predicted accuracy for that candidate.

$$A(v, k, \text{Exploration}) = \sigma\left(\text{MLP}\left(z(v, k, \text{Exploration})\right)\right) \quad (1)$$

This is repeated for all 10 candidates, producing 10 accuracy values. This step is a calculation, not an optimization.

Part B: The optimization problem

We want to:

$$\begin{aligned} &\text{maximize } A(v, k, \text{Exploration}) \\ &\text{minimize } \text{EnergyNorm}(v, k) \\ &\text{minimize } \text{TimeNorm}(v, k) \end{aligned}$$

These three goals are combined into a single score per candidate:

$$\text{CombinedScore}(v, k, \text{Exploration}) = w_A \cdot A(v, k, \text{Exploration}) - w_E \cdot \text{EnergyNorm}(v, k) - w_T \cdot \text{TimeNorm}(v, k) \quad (2)$$

The optimization problem is:

$$\left(v^*(\text{Exploration}), k^*(\text{Exploration})\right) = \arg \max_{(v, k)} \text{CombinedScore}(v, k, \text{Exploration}) \quad (3)$$

$$\begin{aligned} &\text{subject to: } v \in \{\text{none}, \text{small}, \text{medium}, \text{large}\} \\ &\quad k \in \{3, 5, 7\} \text{ if } v \neq \text{none} \\ &\quad w_A + w_E + w_T = 1 \end{aligned}$$

This selects, out of the 10 candidates, the (v, k) pair with the highest CombinedScore, which becomes the verifier and draft length used for Exploration.

3 Reasoning

Part A: Calculating the accuracy term

For each of the 10 candidates (v, k) , the predicted accuracy $A(v, k, \text{Reasoning})$ is produced by the same four-step calculation as Exploration, using the same query as input:

1. The query is converted into numbers by an encoder. Separately, a written description of verifier v is converted into numbers by the same encoder, and the numeric value of k is attached alongside it. These are joined together into one vector, $z(v, k, \text{Reasoning})$.
2. $z(v, k, \text{Reasoning})$ is passed through the MLP, producing one raw number.
3. That raw number is passed through σ , squashing it into the range 0 to 1.
4. The result is $A(v, k, \text{Reasoning})$, the predicted accuracy for that candidate.

$$A(v, k, \text{Reasoning}) = \sigma\left(\text{MLP}\left(z(v, k, \text{Reasoning})\right)\right) \quad (4)$$

This is a calculation, not an optimization.

Part B: The optimization problem

We want to:

$$\begin{aligned} &\text{maximize } A(v, k, \text{Reasoning}) \\ &\text{minimize } \text{EnergyNorm}(v, k) \\ &\text{minimize } \text{TimeNorm}(v, k) \end{aligned}$$

$\text{EnergyNorm}(v, k)$ and $\text{TimeNorm}(v, k)$ are unchanged from Exploration, the same offline reference table values, since they don't depend on the task.

These three goals are combined into a single score per candidate:

$$\text{CombinedScore}(v, k, \text{Reasoning}) = w_A \cdot A(v, k, \text{Reasoning}) - w_E \cdot \text{EnergyNorm}(v, k) - w_T \cdot \text{TimeNorm}(v, k) \quad (5)$$

The optimization problem is:

$$\left(v^*(\text{Reasoning}), k^*(\text{Reasoning})\right) = \arg \max_{(v, k)} \text{CombinedScore}(v, k, \text{Reasoning}) \quad (6)$$

$$\begin{aligned} &\text{subject to: } v \in \{\text{none}, \text{small}, \text{medium}, \text{large}\} \\ &\quad k \in \{3, 5, 7\} \text{ if } v \neq \text{none} \\ &\quad w_A + w_E + w_T = 1 \end{aligned}$$

This selects, out of the 10 candidates, the (v, k) pair with the highest CombinedScore , which becomes the verifier and draft length used for Reasoning.

4 Conclusion

Part A: Calculating the accuracy term

For each of the 10 candidates (v, k) , the predicted accuracy $A(v, k, \text{Conclusion})$ is produced by the same four-step calculation as Exploration and Reasoning, using the same query as input:

1. The query is converted into numbers by an encoder. Separately, a written description of verifier v is converted into numbers by the same encoder, and the numeric value of k is attached alongside it. These are joined together into one vector, $z(v, k, \text{Conclusion})$.
2. $z(v, k, \text{Conclusion})$ is passed through the MLP, producing one raw number.
3. That raw number is passed through σ , squashing it into the range 0 to 1.
4. The result is $A(v, k, \text{Conclusion})$, the predicted accuracy for that candidate.

$$A(v, k, \text{Conclusion}) = \sigma\left(\text{MLP}\left(z(v, k, \text{Conclusion})\right)\right) \quad (7)$$

This is a calculation, not an optimization.

Part B: The optimization problem

We want to:

$$\begin{aligned} &\text{maximize } A(v, k, \text{Conclusion}) \\ &\text{minimize } \text{EnergyNorm}(v, k) \\ &\text{minimize } \text{TimeNorm}(v, k) \end{aligned}$$

$\text{EnergyNorm}(v, k)$ and $\text{TimeNorm}(v, k)$ are unchanged from Exploration and Reasoning, the same offline reference table values.

These three goals are combined into a single score per candidate:

$$\text{CombinedScore}(v, k, \text{Conclusion}) = w_A \cdot A(v, k, \text{Conclusion}) - w_E \cdot \text{EnergyNorm}(v, k) - w_T \cdot \text{TimeNorm}(v, k) \quad (8)$$

The optimization problem is:

$$\left(v^*(\text{Conclusion}), k^*(\text{Conclusion})\right) = \arg \max_{(v, k)} \text{CombinedScore}(v, k, \text{Conclusion}) \quad (9)$$

$$\begin{aligned} &\text{subject to: } v \in \{\text{none}, \text{small}, \text{medium}, \text{large}\} \\ &\quad k \in \{3, 5, 7\} \text{ if } v \neq \text{none} \\ &\quad w_A + w_E + w_T = 1 \end{aligned}$$

This selects, out of the 10 candidates, the (v, k) pair with the highest CombinedScore, which becomes the verifier and draft length used for Conclusion.

5 Summarizing optimization problem, all three tasks

Each task’s optimization problem is solved independently, giving three winning candidates: $(v^*(\text{Exploration}), k^*(\text{Exploration}))$, $(v^*(\text{Reasoning}), k^*(\text{Reasoning}))$, $(v^*(\text{Conclusion}), k^*(\text{Conclusion}))$.

The overall objective for one full query, combining all three tasks, is the sum of the three winning CombinedScores:

$$\max_{(v,k,s)} \sum_{s \in \{\text{Exploration}, \text{Reasoning}, \text{Conclusion}\}} \text{CombinedScore}(v^*(s), k^*(s), s) \quad (10)$$

Since the three tasks place no constraints on each other (the choice made for Exploration does not restrict the choices available for Reasoning or Conclusion), this summed objective is maximized by solving each of the three per-task argmax problems independently and summing their resulting scores:

$$\begin{aligned} \sum_s \text{CombinedScore}(v^*(s), k^*(s), s) = & \text{CombinedScore}(v^*(\text{Exploration}), k^*(\text{Exploration}), \text{Exploration}) \\ & + \text{CombinedScore}(v^*(\text{Reasoning}), k^*(\text{Reasoning}), \text{Reasoning}) \\ & + \text{CombinedScore}(v^*(\text{Conclusion}), k^*(\text{Conclusion}), \text{Conclusion}) \end{aligned} \quad (11)$$

$$\begin{aligned} \text{subject to: for each } s, \quad & v^*(s) \in \{\text{none}, \text{small}, \text{medium}, \text{large}\} \\ & k^*(s) \in \{3, 5, 7\} \text{ if } v^*(s) \neq \text{none} \\ & w_A + w_E + w_T = 1 \quad (\text{same weights across all three tasks}) \end{aligned}$$

This is the generalised optimization problem for DVR’s routing: three independent per-task selections, each maximizing accuracy while minimizing energy and time, summed to describe the total combined score achieved across one full query’s response.